Integrated Climate-intelligent Salt Production Ecosystem: A Multi-modal AI Framework For Sustainable Yield Optimization In Tropical Coastal Environments

Project ID: 25-26J-431

Supervised by: Dr. Nathali Silva
Co-supervised by: Mrs. Thilini Jayalath

Agenda

1. Problem & motivation
2. Proposed solution overview (architecture)
3. System Architecture & Integration
4. Commercialization
5. Demo Plan + roles

Problem Statement & Stakeholders

Context:

Sri Lanka's traditional solar salt industry (Puttalam Salt Society)
10,000+ salterns operating without any digital infrastructure.

Stakeholders:

- Primary - Salt landowners, Salt distributors, Quality testing labs
- Secondary - Salt society (PSS)
- Decision makers - Salt landowners, Salt distributors, Quality testing labs

Observed issue:

- Unpredictable yields
- subjective quality grading
- no supply chain visibility
- unmanaged waste.

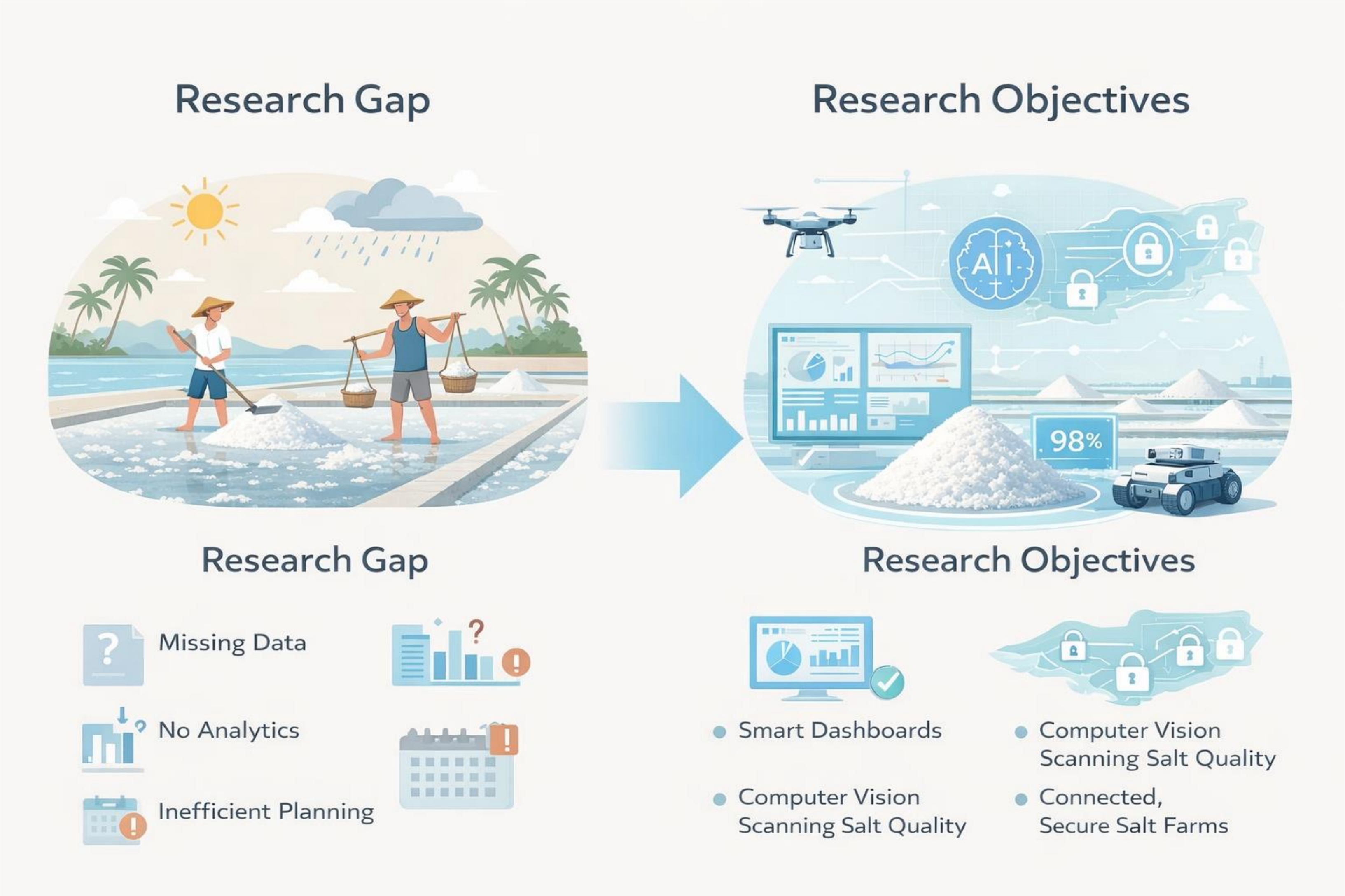
Why it matters:

- Economic losses
- Buyer disputes
- Zero digital traceability.

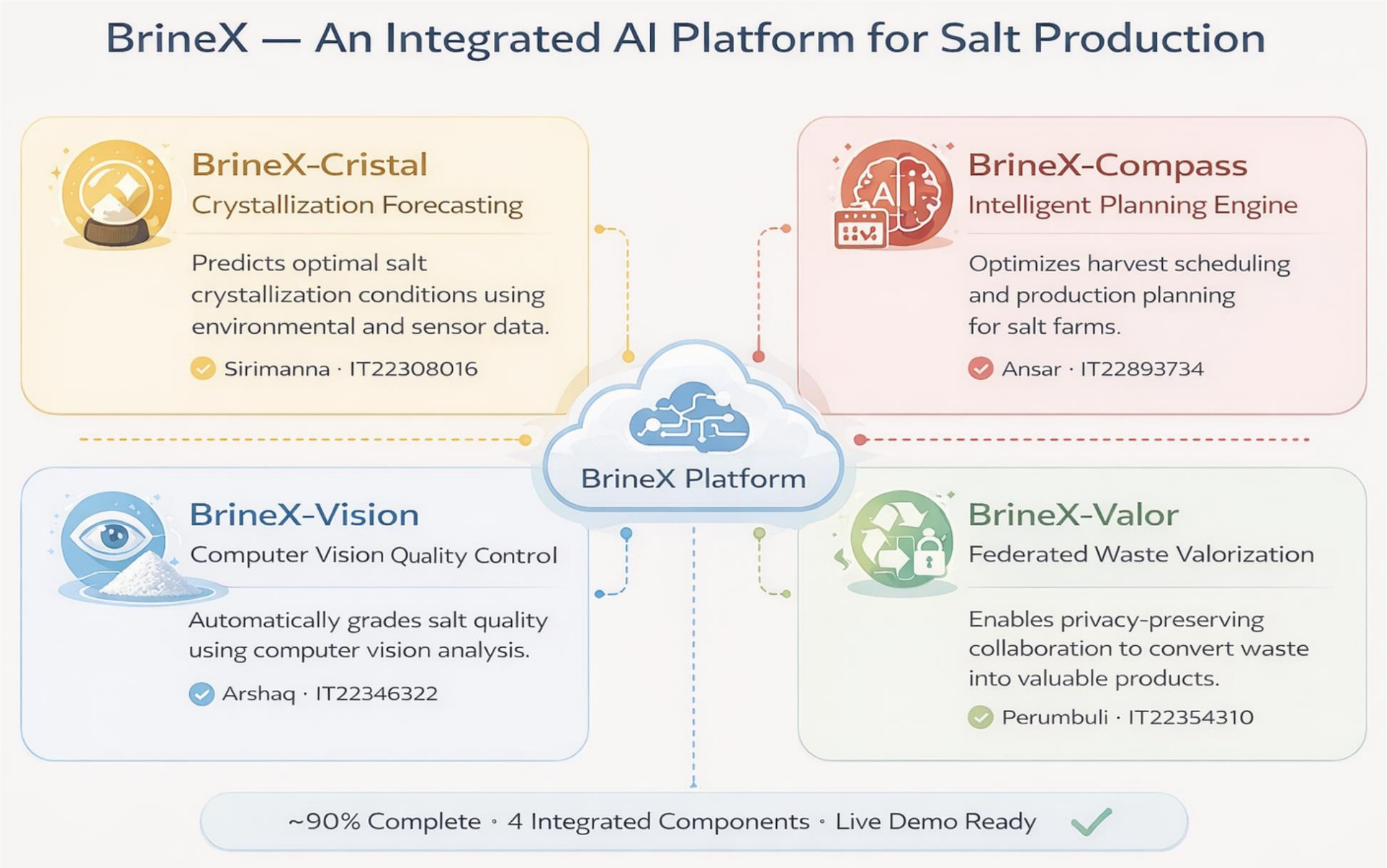
Research Gap & Objectives

Gap: AI applied in agriculture & fisheries industries, never in salt production.
No domain-specific forecasting, planning, or quality tools exist.

Objective: Integrated, modular AI platform purpose-built for tropical coastal salt production.



Solution Overview



← Functional Aspect

Non Functional :

- Real-time detection
- Mobile-responsiveness
- Data privacy (zero raw data shared in Federated Architecture)
- Multilingual UI.

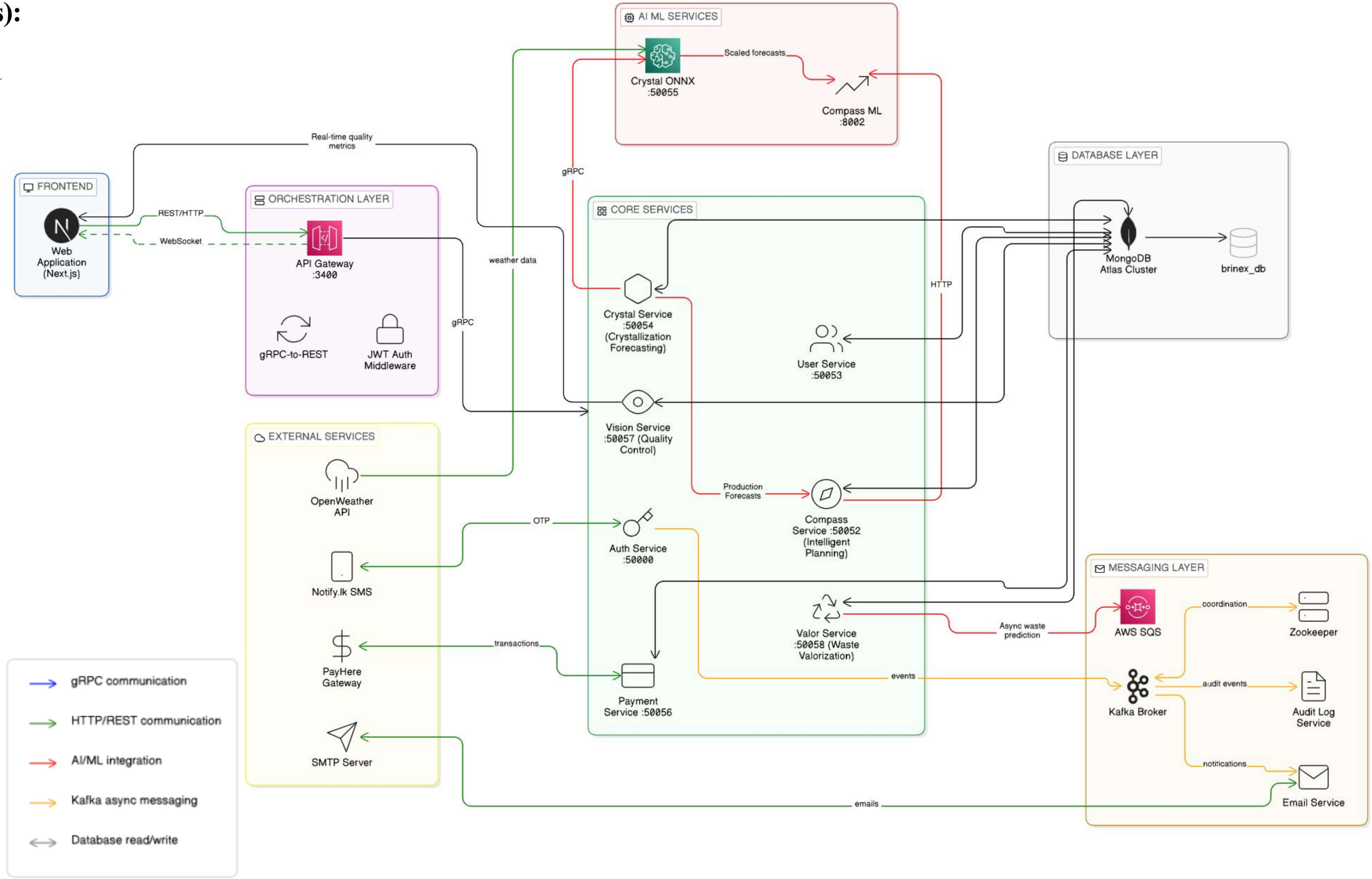
System Architecture & Integration

Integration points (APIs/data formats):

- Crystal → Compass:
  Monthly production forecasts (gRPC, JSON bags/season)
- Crystal → Valor:
  Production volume data (gRPC , JSON kg/month)
- OpenWeather → Crystal:
  Weather forecast (REST , JSON temp/rain/humidity)
- All services → API Gateway
  (gRPC → REST , JWT Bearer)

Deployment:

- Single-machine Docker setup for demo
- Cloud-ready (AWS EC2/ECS) for production.



Commercialization / Impact

Target Market: Puttalam Salterns
→ scaling to Hambantota
→ Global salt market.

Value Proposition: *"Replacing guesswork with AI - from harvest to quality to trade."*

Revenue Model:
Breakeven at just 20 Pro subscribers (~59,980 LKR/month operating cost).
1M LKR/month projected by Year 3.

Adoption Plan:
- Phase 1: Pilot with PSS
- Phase 2: Regional paid rollout
- Phase 3: National scale
- Phase 4: Global Scale.

Free
Rs 0 /forever

For individual landowners getting started with smart salt production.

- Basic crystallization alerts
- 7-day weather integration
- Single saltern monitoring
- Community support
- Basic production logs

Get Started Free

For Landowner

Pro (MOST POPULAR)
Rs 2999 /per month

Full Crystal + Compass access for commercial salt producers scaling operations.

- Advanced crystallization prediction
- 30-day forecasting window
- Unlimited salterns
- Season planning dashboards
- Production analytics
- Priority email support
- Export data & reports

Start Pro Trial

For Landowner, Distributor

Lab
Rs 4999 /per month

Everything in Pro plus Vision AI for laboratories and quality-focused operations.

- Vision AI salt detection
- Purity classification reports
- Live camera integration
- Batch quality management
- Dedicated support
- API access

Start Lab Plan

For Laboratory

Demo Plan

Member	Component	What They Show
Sirimanna RDIB (IT22308016)	Crystal	Daily measurements, LSTM Parameter Forecast, Production predictions with linear regression algorithm
Ansar TD (IT22893734)	Compass	Landowner dashboard, distributor offers, Demand Predictions, SARIMAX price prediction, deal creation
Arshaq MJM (IT22346322)	Vision	Live YOLOv8 detection, real-time bounding boxes, purity % calculation, batch logging, Real-time report generation
Perumbuli PGRMD (IT22354310)	Valor	Waste prediction, federated learning, privacy preservation, and valorization rate

What to watch for:

- Crystal - Compass forecast integration
- Vision real-time detection
- Federated privacy - 0 bytes raw data shared